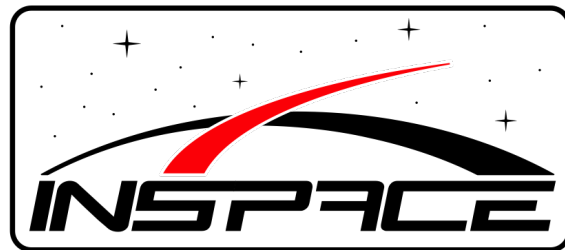


CU InSpace

Radio Packet Format

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1 Introduction

In order to efficiently communicate telemetry data between various devices, Carleton University InSpace has devised its own unique packet format for transmission over LoRa radio. The packet format is not only used for radio communications, but also for logging telemetry data in-flight.

This packet format is made publicly available via the MIT license.

The packet format which follows is made to efficiently transmit flight data recorded by sensors alongside their measurement times. LoRa is a low bandwidth protocol, so it is critical to minimize any unnecessary control information where possible.

2 Packet Layout

CU InSpace's radio packets consist of a packet header followed by one or more segments called blocks. Each block has its own header followed by a variable amount of data determined by the block type. The layout of a packet is shown in figure 2.1.

All fields in this specification are little endian.

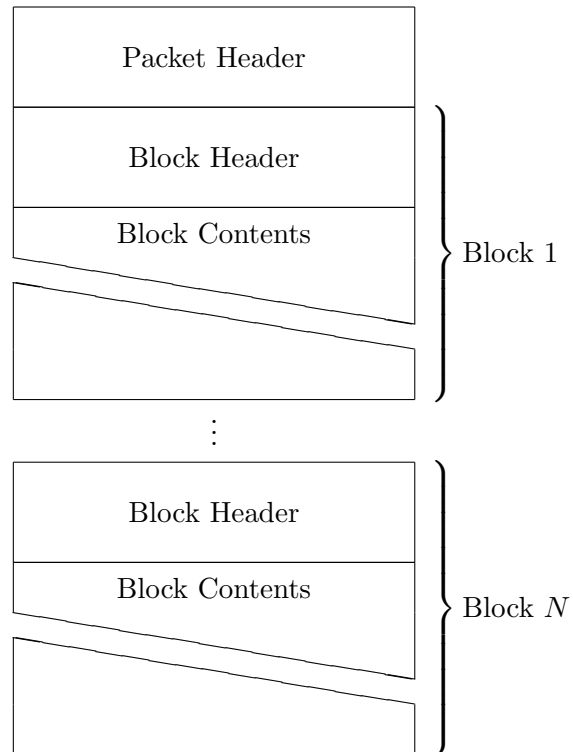


Figure 2.1: Layout of radio packet

2.1 Packet Header

Every packet is preceded by a 13 byte header. The header contains an amateur radio call sign and other information which describes the packet as a whole. The packet header follows the format described in figure 2.2.

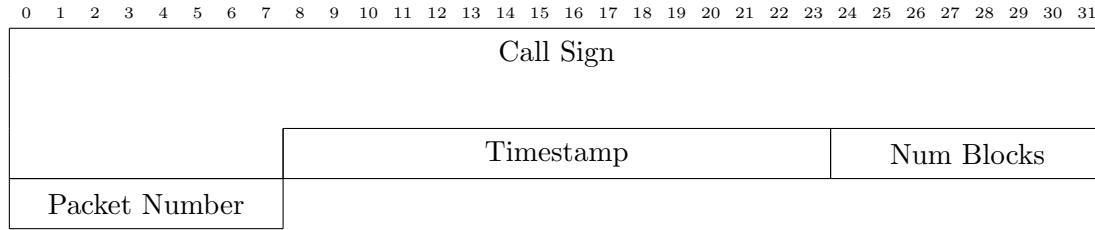


Figure 2.2: Packet header format

Call Sign The first 9 bytes of the packet must contain an amateur radio call sign in ASCII. If the call sign being used is less than 9 bytes long, the call sign will be padded with NULL characters at the end.

The call sign field is 9 bytes long because this field is required to accommodate more than just a single Canadian call sign of maximum 6 characters, but also Canadian call signs being used in the United States. Such call signs must be suffixed with the call zone in which they are being used. [1]

For example, the Canadian amateur call sign of VA3INI would become VA3INI/W5, where W5 is the 5th call zone (coincidentally the call zone for the location where Spaceport America Cup is hosted).

There are 10 call zones in the United States, represented using digits 0-9. [2] The digit is preceded by a forward slash and a capital 'W', for a total of three additional characters on top of the original 6 for a Canadian HAM radio call sign.

Timestamp This field is an unsigned 16 bit integer representing a number of half-minutes since power on. This timestamp serves as a base timestamp to which measurement timestamp offsets are added to.

For instance, the packet header may contain a value of 1 in this field, indicating the base timestamp is 30 seconds since power on. A subsequent block may be a temperature block with a measurement timestamp field containing the value of +12 milliseconds. This value is to be added to the packet header field, meaning that the temperature measurement was taken at 30 seconds and 12 milliseconds after power on.

This schema allows millisecond resolution on measurement time, while preserving a maximum mission duration of around 546 hours, as seen in equation 1

$$\frac{2^{16} - 1}{2 * 60} = 546.125 \quad (1)$$

Num Blocks This field is an unsigned 8 bit integer that contains the total number of blocks in the packet.

Packet Number This field is an unsigned 8 bit integer. It is a rolling counter of the packet sequence number, starting at 0 and increasing until 255 before rolling over again.



Reading this field off of received packets should give an indication of whether any packets were lost in transmission.

2.2 Block Header

Every block starts with a block header. This header indicates the type of information contained in the subsequent block, which allows for parsing of the received data. The block header is formatted as described in Figure 2.3.

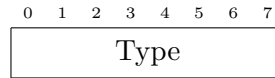


Figure 2.3: Block header format

Type The block type field indicates the information contained in the block. Possible values are listed in table 2.1. Block types are discussed in more detail in section 3.

Block type	Value
Altitude above sea level	0x00
Altitude above launch level	0x01
Temperature	0x02
Pressure	0x03
Linear acceleration	0x04
Angular velocity	0x05
Humidity	0x06
Coordinates (latitude and longitude)	0x07
Voltage	0x08
Magnetic field	0x09
Status message	0x0A
Error message	0x0B
Reserved for future use	0x0C through 0xFF

Table 2.1: Block types

3 Block Formats

Blocks have a type which indicates their contents. The possible block types are listed in table 2.1. This section describes the possible block types and their associated data formats.

3.1 Altitude

The altitude block is used to convey the altitude of the rocket. The format of the altitude block payload is described in Figure 3.1.

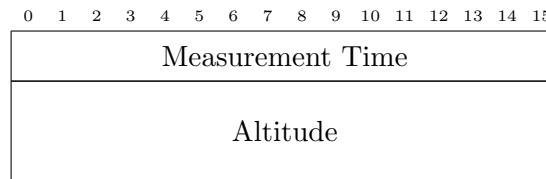


Figure 3.1: Altitude Data Payload Format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

Altitude The calculated altitude in units of millimetres. This field is a signed 32 bit integer in two's complement format.

If the block header specifies the type for altitude above sea level, this measurement should be interpreted as millimetres above sea level.

If the block header specifies the type for altitude above launch height (ground level), this measurement should be interpreted as millimetres from the starting height of the rocket.

See Table 2.1 for the different altitude types.

3.2 Linear Acceleration

The linear acceleration block is used to convey generic 3-axis accelerometer data. This block is intended to abstract over the details of any individual accelerometer. The format of the acceleration block payload is described in figure 3.2.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Measurement Time															
X-Axis															
Y-Axis															
Z-Axis															

Figure 3.2: Acceleration Data Payload Format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

***-Axis** These fields represent the acceleration measurements for each axis. These fields are signed integers in two's complement format.

The unit of measurement for each axis is measured in centimetres per second squared. That is, dividing the value in each axis field by 100 will give the standard acceleration unit of metres per second squared.

The acceleration measurements are relative to the rocket's heading. If the IMU Z-axis is in parallel with the rocket, then the acceleration in the z-axis is also a parallel vector.

3.3 Angular Velocity

The angular velocity block is used to convey generic 3-axis gyroscope data. This block is intended to abstract over the details of any individual gyroscope. The format of the acceleration block payload is described in figure 3.3.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Measurement Time															
X-Axis															
Y-Axis															
Z-Axis															

Figure 3.3: Angular Velocity Data Payload Format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

***-Axis** These fields represent the angular velocity measurements for each axis. These fields are signed integers in two's complement format.

Each field describes angular velocity in tenths of degrees per second. In other words, dividing this field by 10 will give you angular rotation in a standard unit of degrees per second.

3.4 Coordinates

The coordinates block is used to convey the geographical (latitude and longitude) coordinates of the rocket. The format of the coordinates block is described in figure 3.4.

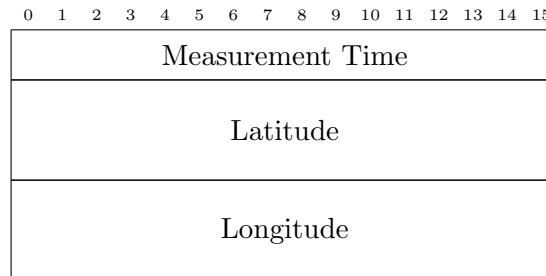


Figure 3.4: Coordinate Data Payload Format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

Latitude Measured in tenths of micro-degrees. This field is a signed 32 bit integer. To convert to degrees, divide by 10^7 .

Longitude Measured in tenths of micro-degrees. This field is a signed 32 bit integer. To convert to degrees, divide by 10^7 .

3.5 Humidity

The humidity block is used to convey the relative humidity inside of the rocket. The format of the humidity block is described in figure 3.5.

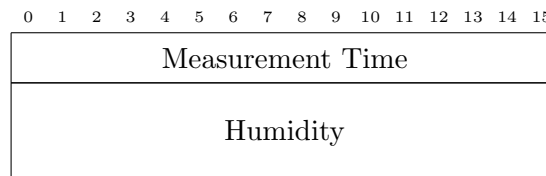


Figure 3.5: Humidity Data Payload Format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.



Humidity The calculated relative humidity in ten thousandths of a percent (i.e 1% is 100). This field is an unsigned 32 bit integer.

3.6 Pressure

The pressure data block reports the pressure detected from within the rocket in Pascals. The format of the pressure block payload is described in figure 3.6.

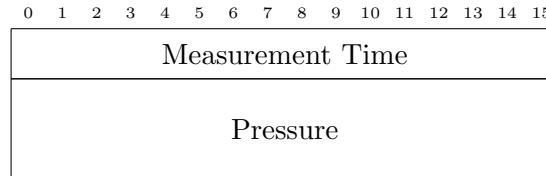


Figure 3.6: Pressure data block format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

Pressure The measured pressure in units of Pascals. This field is an unsigned 32 bit integer.

3.7 Temperature

The temperature data block reports the temperature detected from within the rocket in millidegrees Celsius. The format of the temperature block payload is described in figure 3.7.

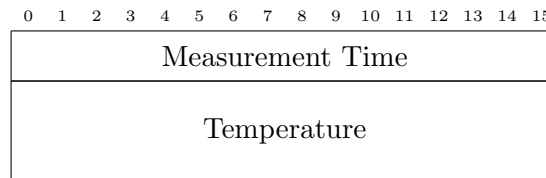


Figure 3.7: Temperature data block format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

Temperature The measured temperature in units of millidegrees Celsius. This field is a signed 32 bit integer in two's complement format.

3.8 Voltage

The voltage data block reports the voltage measured on a specific electrical trace within the rocket. The measurement is associated with a generic numerical ID for identification by the receiver. This allows multiple voltage points to be measured within the rocket. The format of the voltage block payload is described in figure 3.8.

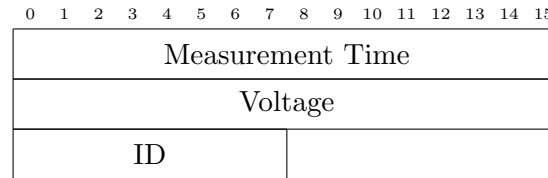


Figure 3.8: Voltage data block format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

Voltage The measured voltage in units of millivolts. This field is a signed 16 bit integer.

ID A numerical identifier associated with the voltage measurement for identification by the receiver. This field is an unsigned 8 bit integer, allowing for up to 255 unique identifiers.

3.9 3-Axis Magnetic Field

The magnetic field block is used to convey generic 3-axis magnetometer data. This block is intended to abstract over the details of any individual magnetometer. The format of the 3-axis magnetic field block payload is described in figure 3.9.

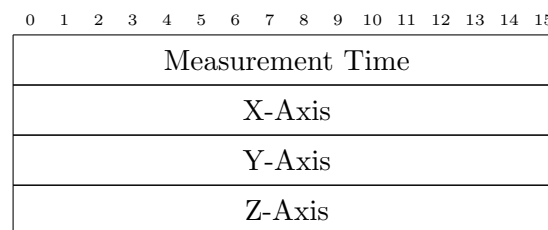


Figure 3.9: 3-Axis Magnetic Field Data Payload Format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

***-Axis** These fields represent the magnetic field measurements for each axis. These fields are signed integers in two's complement format.

The unit of measurement for each axis is 0.1 microtesla.

The magnetic field measurements are relative to the rocket's heading. If the magnetometer Z-axis is parallel with the rocket, then the magnetic field in the z-axis is also a parallel vector.

3.10 Status Message

The status message block type reports information about the current state of the telemetry system in integer codes, such as the current flight state or flight events that are detected. The format of the status message payload is described in figure 3.10.

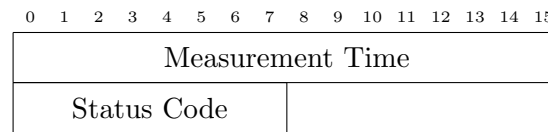


Figure 3.10: Status message data block format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

Status Code A status code is an integer representing information about the rocket's state, the state of the telemetry system, or other information. The measurement time describes a time that the status code was valid, such as when a state change or event occurs, or any time afterwards. The same code can be transmitted multiple times to ensure that a receiver gets has an up to date picture of the telemetry system's operation. For example, the state of the telemetry system could be transmitted periodically. The possible status codes are described in 3.1.

Status message	Value
Systems nominal	0x00
Telemetry in idle state	0x01
Telemetry in airborne state	0x02
Telemetry in landed state	0x03
Rocket ascent detected	0x04
Rocket apogee detected	0x05
Rocket descent detected	0x06
Reserved for future use	0x07 through 0xFF

Table 3.1: Status message codes



3.11 Error Message

The error message block type reports information about failures in the telemetry system. The format of the error message payload is described in figure 3.11.

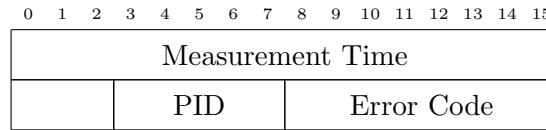


Figure 3.11: Error message data block format

Measurement Time A signed 16 bit integer in units of milliseconds. This timestamp is meant to be added as an offset to the base timestamp in the packet header. This is explained in section 2.1.

PID The PID (process ID) or originating process is a number that corresponds to the context that the error occurred in, denoting which process, thread, or hardware component the error occurred in. A value of 0x00 represents a general error, not specific to any context. Other values are left to the telemetry system to define.

Error Codes The error codes describe the specific type of error that has occurred. The value of 0x00 is defined as a general error, with an unknown source. Other values are left to the telemetry system to define.

References

- [1] S. Bertuzzo. “Operating amateur radio in foreign countries,” Accessed: Feb. 11, 2024. [Online]. Available: <https://www.rac.ca/operating/operation-in-foreign-countries/>.
- [2] S. Stroobandt. “Us call areas & zones by state,” Accessed: Feb. 11, 2024. [Online]. Available: <https://hamwaves.com/map.us/en/index.html>.